Edward Y. Chang
Associate Professor
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University of California
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EDUCATION

Stanford University, Computer Science, M.S.	1992-94
Stanford University, Electrical Engineering, Ph.D.	1995-99

AWARDS

NSF Career Award	2002
IBM Faculty Partnership Award	2000-2002

EXPERIENCE

Associate Professor, University of California, Santa Barbara	2003-present
Assistant Professor, University of California, Santa Barbara	1999-2003
Staff Engineer, Sun Microsystems, Mountain View, CA	1994-95

SELECTED PATENTS

"*Exploiting Geometric Properties for SVM Indexing*," with University of California., 2004.
"*Kernel Alignment for Learning Imbalance Datasets*," with University of California., 2003.
"*DPF: Dynamic Partial Function*," with VIMA Technologies Inc., 2002.
"*Perception-Based and Personalizable Image Retrieval*," with University of California, 2001.

SYNERGISTIC ACTIVITIES

Associate Editor, for IEEE Transactions on Knowledge and Data Engineering 2003-present
Guest Editor, for ACM Multimedia System Journal Special Issue on Video Surveillance 2004
General Co-Chair, for ACM International Conference on Multimedia 2006
Program Co-Chair, for IS&T/SPIE Conf. on Retrieval Methods for Multimedia 2005/2006
Program Co-Chair, for ACM Workshop on Video Surveillance and Sensor Networks 2003/2004
Program Co-Chair, for IEEE Workshop on Emerging Multimedia Applications (with ICDE) 2005
Co-Chair, for ARDA Workshop on Large Scale Ontology for Multimedia Understanding 2004
Associated PC Chair, for ACM International Conference on Multimedia 2002/2003/2004
PC Area Chair, for IEEE Multimedia Conference 2004/2005
PC Member, for ACM SIGMOD 2004, SIAM Data Mining 2004, ACM CIKM 2003/2004, IEEE MM 2002/2003/2004/2005, WISE 2002/2003/2004/2005, ICCV 2003, ICDE 2002, etc.

SELECTED INVITED TALKS

- "*Event Sensing on Distributed Video-Sensor Networks*," Keynote, Basenets 2004, in cooperation with ACM/IEEE Conf. on Broadband Networks, San Jose, October 2004.
- "*Perception-based Image Retrieval*,", Google, January 2004.
- "*Multimedia Semantics and Machine Learning*," Tutorial, International Conf. on Multimedia, Baltimore, July 2003.
- "*Statistical Learning Methods for Emerging Database Applications*," Plenary Tutorial, International Conf. on Database Systems for Advanced Applications, Kyoto, March 2003.
- "*Multimedia Information Retrieval*" Panel, International Conf. on Multimedia, Sept. 2002.
- "*Multimedia Data Mining*," HP Labs., December 2001; January 2002.
- "*Dynamic Partial Function*," UC Berkeley Digital Library Seminar, February 2002.
- "*Measuring and Learning Perceptual Similarity*," UCLA, October 2001.
- "*Measuring and Learning Perceptual Similarity*," INRIA-Rocquencourt, September 2001.

TEN SELECTED PUBLICATIONS (ALL RELATED)

- "*Enhanced Perceptual Distance Functions and Indexing for Image Replica Recognition*," A. Qamra, Y. Meng, and E. Y. Chang, IEEE Transactions on Pattern Analysis and Machine Intelligence, Volume 27, Number 3, March 2005.
- "*KBA: Kernel Boundary Alignment Considering Imbalanced Data Distribution*," G. Wu and E. Y. Chang, IEEE Transactions on Knowledge and Data Engineering, 2005 (to appear).
- "*Anatomy of a Multi-camera Security Surveillance System*," L. Jiao, Y. Wu, G. Wu, E. Y. Chang, Y.F. Wang, ACM Multimedia System Journal Special Issue, Volume 10, Number 2, pp. 144-163, November 2004.
- "*Concept-dependent Multimodal Active Learning for Image Retrieval*," K. Goh and E. Y. Chang, the 12^{th} ACM International Conf. on Multimedia, pp.564-571, October 2004.
- "*Adaptive Stream Resource Management Using Kalman Filters*," A. Jain, E. Y. Chang, and Y.F. Wang, ACM International Conference on Management of Data (SIGMOD), June 2004.
- "*Confidence-based Dynamic Ensemble for Image Annotation and Semantics Discovery*," (best student paper finalists), B. Li, K.-S. Goh, and E. Chang, the 11^{th} ACM International Conference on Multimedia, pp. 195-206, November 2003.
- "*The Discovery of A Perceptual Distance Function for Measuring Image Similarity*," B. Li, E. Chang, and Y. Wu, ACM Multimedia Journal Special Issue on Content-Based Image Retrieval, Volume 8, Number 6, pp. 512-522, April 2003.
- "*DynDex: A Dynamic and Non-metric Space Indexer*," K. Goh, B. Li, and E. Chang, in Proc. of the 10^{th} ACM International Conf. on Multimedia, pp.466-475, December 2002.
- "*Approximate Similarity Queries in High-Dimensional Spaces*," C. Li, E. Chang, H. Garcia-Molina, and G. Wiederhold, IEEE Transactions on Knowledge and Data Engineering, Volume 14, Number 4, pp.792-808, July/August 2002.
- "*Support Vector Machine Active Learning For Image Retrieval*," S. Tong and E. Chang, in Proc. of the 9^{th} ACM International Conf. on Multimedia, pp.107-118, October 2001.

SELECTED INVITED PAPERS

- "*Multimodal Kernel Fusion for News Video Concept Detection*," Y. Wu, C.Y. Lin, E. Y. Chang and J. R. Smith, IEEE International Conference on Image Processing, October 2004.
- "*Active Learning and its Scalability for Image Retrieval*," E. Y. Chang and W.C. Lai, IEEE International Conference on Multimedia, June 2004.
- "*Statistical Learning for Effective Visual Image Retrieval*," E. Chang, B. Li, and G. Wu, IEEE International Conf. on Image Processing, September 2003.
- "*On Learning Perceptual Distance Functions for Image Retrieval*," B. Li and E. Chang, in Proc. of IEEE International Conf. on Acoustics, Speech and Signal Processing, May 2002.
- "*High-dimensional Visual Data Indexing*," K. Goh and E. Chang, to appear in Proc. of the 6^{th} International Conf. on Visual Databases, May 2002.
- "*Learning and Measuring Perceptual Similarity*," E. Chang, in Proc. of the 1^{st} NSF/INRIA/Berkeley/IBM Workshop on Content-based Multimedia Information Retrieval, September 2001.

COLLABORATORS & GRADUATE STUDENTS

Graduate Advisor: Hector Garci-Molina;
Collaborators: Chen Li, Simon Tong, James Wang, Yuan-Fang Wang, Gio Wiederhold;
Graduate Advisees: Zoran Dimitrijevic, Kingshy Goh, Ankur Jain, Beitao Li, Yan Meng, Navneet Panda, Arun Qamra, Raju Rangaswami, Gerard Sychay, Gang Wu, Yi Wu, Huaxin You.